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MATHEMATICAL
WORKS
OF ÉVARISTE GALOIS,

PUBLISHED
UNDER THE AUSPICES OF THE MATHEMATICAL SOCIETY OF
FRANCE,

WITH AN INTRODUCTION

BY

M. ÉMILE PICARD,
MEMBER OF THE INSTITUTE OF FRANCE.



PARIS,
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MATHEMATICAL
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55, quai des Grands-Augustins.



E. GALOIS



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INTRODUCTION.

The works of Galois were, as is well known, published in 1846 by Liouville in the *Journal de Mathématiques*. It was regrettable that the works of the great geometer could not be obtained as a separate volume; the Mathematical Society therefore decided to have Galois's memoirs reprinted. This edition conforms to the preceding one; only the notice placed by Liouville at the beginning of the publication has been omitted.

A study of Galois's life, apparently definitive, has just been published by M. Paul Dupuy in the *Annales de l'École Normale supérieure* (1896). Among the earlier documents concerning Galois's life, mention must be made of the obituary notice devoted to him by his friend Auguste Chevalier in the *Revue encyclopédique* (September 1832), and of an article published in the *Magasin pittoresque* in 1848. Évariste Galois was born at Bourg-la-Reine, near Paris, on 25 October 1811; he left his father's home in 1823 to enter the fourth form at the Collège Louis-le-Grand. From the age of fifteen his extraordinary aptitude for the mathematical sciences began to show itself. Elementary algebra books did not satisfy him, and he acquired his algebraic education from Lagrange's classic works. It appears that by the age of seventeen Galois had already obtained results of the highest importance concerning the theory of algebraic equations. We can do no more than

conjecture about the course of his ideas, since the two memoirs he submitted to the Academy of Sciences were lost. One thing, however, is certain: at the beginning of 1830 he possessed his general principles, as is shown by the summary of a memoir on the algebraic solution of equations in the *Bulletin de Férussac*. It states a series of results that are plainly applications of a general theory. This short article is the most important one published by Galois himself, for the fundamental memoir on algebra found among his papers was not printed until 1846.

M. Dupuy's article contains information of great interest about Galois's life. It is unlikely that new documents will now be added to those we possess. After twice failing the entrance examination for the École Polytechnique, Galois entered the École Normale in 1829 and was compelled to leave it the following year. During the last year of his life he devoted himself entirely to politics, spent several months behind the bars of Sainte-Pélagie, and, mortally wounded in a duel, died on 31 May 1832. Confronted with a life so short and so troubled, our admiration redoubles for the prodigious genius who left so deep a mark on science. Precocious achievement is not rare among great geometers, but Galois's case is remarkable above all others. Alas, the unhappy young man seems to have paid a bitter price for his genius. As his brilliant mathematical faculties develop, his formerly cheerful and open character darkens, and the consciousness of his immense superiority gives rise to excessive pride. This caused the disappointments that so strongly affected his career, the first being his failure at the École Polytechnique. His examination at that

school left vivid memories. Without going as far as legend would have us go, let us say only that Galois refused to answer a question, which he considered ridiculous, about the arithmetic theory of logarithms. Nor can there be any doubt that he was unwilling to provide the explanations of his work requested by the mathematicians with whom he came into contact—explanations made necessary by the hurried composition of his memoirs. It is therefore easy to understand why his merit was not recognized by his contemporaries. Liouville did not succeed without difficulty in grasping the sequence of Galois's ideas, and numerous commentators were still needed to fill gaps remaining in more than one proof and to bring the great geometer's theories to the degree of simplicity they are capable of assuming today.

The theory of equations owes considerable advances to Lagrange, Gauss, and Abel, but none of them succeeded in bringing to light the fundamental element on which all the properties of an equation depend. That glory was reserved for Galois, who showed that every algebraic equation has a corresponding group of substitutions in which the equation's essential characteristics are reflected. In algebra, group theory had previously been the subject of many investigations, mostly by Cauchy, who had already introduced certain elements of classification. Galois's study of the theory of equations showed him the importance of the notion of an invariant subgroup of a given group. He was thus led to divide groups into simple and composite groups, a fundamental distinction which in fact extends far beyond algebra to the concept of groups of operations in its broadest sense.

For general theories to acquire in science a

permanent right of citizenship, they usually need to be illustrated by particular applications. In several fields such applications are not always easy to find, and one could cite more than one theory in modern mathematics confined, if I may say so, within its excessive generality. From the artistic point of view, which plays a vital role in pure mathematics, nothing is more satisfying than the development of these great theories; nevertheless, thoughtful minds regret this tendency, which may have its dangers. No such regret can be expressed in Galois's case. From the beginning, the algebraic solution of equations provided him with a particular field of application in which many geometers later followed him, foremost among them M. Camille Jordan.

Galois's work on algebraic equations made his name famous, but he appears to have made discoveries in analysis that were at least as important. In his letter to Auguste Chevalier, written on the eve of his death and constituting a kind of scientific testament, Galois speaks of a memoir that could be composed from his researches on integrals. We know of those researches only what he says in this letter. Several points in some of Galois's statements remain obscure; nevertheless, one can form an exact idea of some of the results he had reached in the theory of integrals of algebraic functions. One thereby becomes convinced that he possessed the most essential results concerning Abelian integrals that Riemann was to obtain twenty-five years later. It is with astonishment that we see Galois speak of the periods of an Abelian integral associated with an arbitrary algebraic function. For hyperelliptic integrals we have no difficulty understanding what he means by *period*, but the general case is different, and

one is almost tempted to suppose that Galois had at least glimpsed certain notions concerning functions of a complex variable that were not to be developed until several years after his death. The statements are precise: the illustrious author classifies Abelian integrals into three kinds and asserts that, if n denotes the number of linearly independent integrals of the first kind, there will be $2n$ periods. The theorem concerning inversion of the parameter and argument in integrals of the third kind is clearly indicated, as are the relations among the periods of Abelian integrals. Galois also speaks of a generalization of Legendre's classical equation involving the periods of elliptic integrals, a generalization that had probably led him to the important relations since discovered by Weierstrass and M. Fuchs. We have said enough to show the scope of Galois's discoveries in analysis. Had a few more years been granted him to develop his ideas in this direction, he would have been Abel's glorious successor and would have constructed, in its essential parts, the theory of algebraic functions of one variable as we know it today. Galois's reflections extended still farther. He ends his letter by speaking of applying the theory of ambiguity to transcendental analysis. One can roughly guess what he means, and in this field, which, as he says, is immense, many discoveries still remain to be made even today.

It is not without emotion that one finishes reading the scientific testament of this twenty-year-old, written on the eve of the day when he was to disappear in an obscure quarrel. His death was an immense loss to science. Had Galois lived, his influence would have greatly modi-

fied the direction of mathematical research in our country. I shall not risk hazardous comparisons: Galois undoubtedly has equals among the great mathematicians of this century; none surpasses him in the originality and depth of his conceptions.

ÉMILE PICARD,
President of the Mathematical Society of France.



MATHEMATICAL WORKS OF ÉVARISTE GALOIS.

I. — ARTICLES PUBLISHED BY GALOIS.

PROOF OF A THEOREM ON PERIODIC CONTINUED FRACTIONS⁽¹⁾.

It is known that if, by Lagrange's method, one expands one root of a quadratic equation as a continued fraction, that continued fraction will be periodic. The same is true of one root of an equation of arbitrary degree if that root is a root of a rational quadratic factor of the left-hand side of the given equation; in that case the equation will have at least one other root that is also periodic. In either case the continued fraction may be immediately periodic or not imme-

¹Volume XIX of M. Gergonne's *Annales de Mathématiques*, page 294 (1828–1829). Galois was then a pupil at the Collège Louis-le-Grand. (J. LIOUVILLE.)

diately periodic; but in the latter case there will at least be one transformed equation having a root that is immediately periodic.

Now, when an equation has two periodic roots belonging to the same rational quadratic factor, and one of them is immediately periodic, there is a rather singular relation between these two roots which appears not to have been noticed before and which may be expressed by the following theorem.

THEOREM. — *If one root of an equation of arbitrary degree is an immediately periodic continued fraction, the equation must have another root, likewise periodic, obtained by dividing negative unity by the same periodic continued fraction written in reverse order.*

Proof. — To fix ideas, let us take periods of only four terms, for the uniform course of the calculation shows that the same would hold if we admitted a larger number. Let one root of an equation of arbitrary degree be expressed as follows:

$$x = a + \frac{1}{b + \frac{1}{c + \frac{1}{d + \frac{1}{a + \frac{1}{b + \frac{1}{c + \frac{1}{d + \dots}}}}}}};$$

the quadratic equation to which this root belongs, and which consequently contains its conjugate, will be

$$x = a + \frac{1}{b + \frac{1}{c + \frac{1}{d + \frac{1}{x}}}};$$

and from this we successively obtain

$$a - x = -\frac{1}{b + \frac{1}{c + \frac{1}{d + \frac{1}{x}}}},$$

$$\frac{1}{a - x} = -b + \frac{1}{c + \frac{1}{d + \frac{1}{x}}},$$

$$b + \frac{1}{a - x} = -\frac{1}{c + \frac{1}{d + \frac{1}{x}}},$$

$$\frac{1}{b + \frac{1}{a - x}} = -c + \frac{1}{d + \frac{1}{x}},$$

$$c + \frac{1}{b + \frac{1}{a - x}} = -\frac{1}{d + \frac{1}{x}},$$

$$\frac{1}{c + \frac{1}{b + \frac{1}{a - x}}} = -d + \frac{1}{x},$$

$$d + \frac{1}{c + \frac{1}{b + \frac{1}{a - x}}} = -\frac{1}{x};$$

$$\frac{1}{d + \frac{1}{c + \frac{1}{b + \frac{1}{a - x}}}} = -x,$$

that is,

$$x = -\frac{1}{d + \frac{1}{c + \frac{1}{b + \frac{1}{a - x}}}};$$

this is therefore the quadratic equation that gives the two roots in question. But by repeatedly substituting for x on its right-hand side that same right-hand side, which is indeed its value, it gives

$$x = -\frac{1}{d + \frac{1}{c + \frac{1}{b + \frac{1}{a + \frac{1}{d + \frac{1}{c + \frac{1}{b + \frac{1}{a + \frac{1}{d + \frac{1}{c + \frac{1}{b + \frac{1}{a + \dots}}}}}}}}}}}}};$$

this is therefore the other value of x given by the equation; as one sees, it equals -1 divided by the first value written in reverse order.

In the preceding argument we supposed that the given root was greater than unity; but if we had

$$x = \frac{1}{a + \frac{1}{b + \frac{1}{c + \frac{1}{d + \frac{1}{a + \frac{1}{b + \frac{1}{c + \frac{1}{d + \dots}}}}}}}};$$

we would conclude, for one value of $\frac{1}{x}$,

$$\frac{1}{x} = a + \frac{1}{b + \frac{1}{c + \frac{1}{d + \frac{1}{a + \frac{1}{b + \frac{1}{c + \frac{1}{d + \dots}}}}}}};$$

the other value of $\frac{1}{x}$ would therefore, by the preceding argument, be

$$\frac{1}{x} = -\frac{1}{d + \frac{1}{c + \frac{1}{b + \frac{1}{a + \frac{1}{d + \frac{1}{c + \frac{1}{b + \frac{1}{a + \dots}}}}}}},$$

whence, for the other value of x , we would obtain

$$x = -d + \frac{1}{c + \frac{1}{b + \frac{1}{a + \frac{1}{d + \frac{1}{c + \frac{1}{b + \frac{1}{a + \dots}}}}}}.$$

or

$$x = -1 : \frac{1}{d + \frac{1}{c + \frac{1}{b + \frac{1}{a + \frac{1}{d + \frac{1}{c + \frac{1}{b + \frac{1}{a + \dots}}}}}}}};$$

which falls exactly under our theorem.

Let A be any immediately periodic continued fraction, and let B be the continued fraction obtained by reversing its period. If one root of an equation is $x = A$, it must have another root $x = -\frac{1}{B}$. Now, if A is positive and greater than unity, $-\frac{1}{B}$ will be negative and lie between 0 and -1 ; conversely, if A is negative and lies between 0 and -1 , $-\frac{1}{B}$ will be positive and greater than unity. Thus, when one root of a quadratic equation is an immediately periodic continued fraction greater than unity, the other must lie between 0 and -1 ; conversely, if one lies between 0 and -1 , the other must be positive and greater than unity.

Conversely, one can prove that if one of the two roots of a quadratic equation is positive and greater than unity while the other lies between 0 and -1 , then those roots can be expressed as immediately periodic continued fractions. Indeed, continue to let A be any positive immediately periodic continued fraction greater than unity, and let B be the immediately periodic continued fraction obtained by reversing its period; like A , it too will be positive and greater than unity. The first root of the given equation cannot have the form

$$x = p + \frac{1}{A},$$

for then, by our theorem, the second would have to be

$$x = a + \frac{1}{-\frac{1}{B}} = a - B;$$

now $a - B$ could not lie between 0 and 1 unless the integer part of B were equal to p , in which case the first value would be immediately periodic. Nor could the first value of x be $x = p + \frac{1}{q + \frac{1}{A}}$, for then the other would be

$$x = p + \frac{1}{q - B} \quad \text{or} \quad x = p - \frac{1}{B - q};$$

now, for this value to lie between 0 and -1 , it would first be necessary that $\frac{1}{B - q}$ equal p plus a fraction. Hence $B - q$ would have to be less than unity, requiring B to equal q plus a fraction. Thus q and p would have to equal, respectively, the first two terms of the period belonging to B , or the last two terms of the period belonging to A . Contrary to the hypothesis, the value $x = p + \frac{1}{q + \frac{1}{A}}$ would therefore be immediately periodic. An analogous

argument would prove that the periods cannot be preceded by a larger number of terms not belonging to them.

Thus, when treating a numerical equation by Lagrange's method, we may be sure that no periodic roots are to be expected until we encounter a transformed equation having at least one positive root greater than unity and another lying between 0 and -1 . If the root being pursued is indeed periodic, its period will begin no later than at this transformed equation.

If one root of a quadratic equation is not only immediately periodic but also symmetric—that is, if terms equally distant from the ends of the period are equal—then $B = A$; consequently the two roots

will be A and $-\frac{1}{A}$, and the equation will therefore be

$$Ax^2 - (A^2 - 1)x - A = 0.$$

Conversely, every quadratic equation of the form

$$ax^2 - bx - a = 0$$

has roots that are both immediately periodic and symmetric. Indeed, substituting in turn infinity and -1 for x gives positive results, whereas substituting $x = 1$ and $x = 0$ gives negative results. We first see that the equation has one positive root greater than unity and one negative root between 0 and -1 , and hence that these roots are immediately periodic. Moreover, the equation does not change when x is replaced by $-\frac{1}{x}$. Thus, if A is one of its roots, the other is $-\frac{1}{A}$, so in this case $B = A$.

Let us apply these general results to the quadratic equation

$$3x^2 - 16x + 18 = 0;$$

we first find that it has a positive root between 3 and 4. Setting

$$x = 3 + \frac{1}{y},$$

we obtain the transformed equation

$$3y^2 - 2y - 3 = 0,$$

whose form tells us that the values of y are both immediately periodic and symmetric. Indeed, setting successively

$$y = 1 + \frac{1}{z}, \quad z = 2 + \frac{1}{t}, \quad t = 1 + \frac{1}{u},$$

we obtain the transformed equations

$$2z^2 - 4z - 3 = 0, \quad 3t^2 - 4t - 2 = 0, \quad 3u^2 - 2u - 3 = 0.$$

The identity of the equations in u and y proves that the positive value

of y is

$$y = 1 + \frac{1}{2 + \frac{1}{1 + \frac{1}{1 + \frac{1}{2 + \frac{1}{1 + \dots}}}}}$$

its negative value will therefore be

$$y = -\frac{1}{1 + \frac{1}{2 + \frac{1}{1 + \frac{1}{1 + \frac{1}{2 + \frac{1}{1 + \dots}}}}}}$$

the two values of x will therefore be

$$x = 3 + \frac{1}{1 + \frac{1}{2 + \frac{1}{1 + \frac{1}{1 + \frac{1}{2 + \frac{1}{1 + \dots}}}}}} \quad x = 3 - \frac{1}{1 + \frac{1}{2 + \frac{1}{1 + \frac{1}{1 + \frac{1}{2 + \frac{1}{1 + \dots}}}}}}$$

the latter of which, by the familiar formula

$$p - \frac{1}{q} = p - 1 + \frac{1}{1 + \frac{1}{q-1}},$$

becomes

$$x = 1 + \frac{1}{1 + \frac{1}{1 + \frac{1}{1 + \frac{1}{1 + \frac{1}{2 + \frac{1}{1 + \frac{1}{1 + \frac{1}{2 + \frac{1}{1 + \dots}}}}}}}}}$$

NOTES

ON

SOME POINTS IN ANALYSIS^(*).

§ I. — Proof of a theorem of analysis.

THEOREM. — *Let Fx and fx be any two given functions; for all x and h we shall have*

$$\frac{F(x+h) - Fx}{f(x+h) - fx} = \varphi(k),$$

where φ is a determinate function and k is a quantity between x and $x+h$.

Proof. — Indeed, set

$$\frac{F(x+h) - Fx}{f(x+h) - fx} = P;$$

from which we obtain

$$F(x+h) - Pf(x+h) = Fx - Pfx;$$

which shows that the function $Fx - Pfx$ does not change when x is replaced by $x+h$. It follows that, unless it remains constant between these limits—which could occur only in special cases—this function will have one or more maxima and minima between x and $x+h$. Let k be the value of x corresponding to one of them. We shall plainly have $k = \psi(P)$, where ψ is a determinate function; hence we must also have $P = \varphi(k)$, where φ is another likewise determinate function; this proves the theorem.

From this we may conclude, as a corollary, that the quantity

$$\lim. \frac{F(x+h) - Fx}{f(x+h) - fx} = \varphi(x),$$

for $h = 0$, is necessarily a function of x , which proves, *a priori*, the existence of derived functions.

^(*)M. Gergonne's *Annales de Mathématiques*, volume XXI, page 182 (1830–1831). Owing to a printing error, it reads *Galais*, student at the *École Normale*, instead of *Galois*. (J. LIOUVILLE.)

§ II. — Radius of curvature of curves in space.

The radius of curvature of a curve at any one of its points M is the perpendicular drawn from that point to the intersection of the normal plane at M with the consecutive normal plane, as is readily verified by geometric considerations.

Accordingly, let (x, y, z) be a point of the curve. The normal plane at this point has equation

$$(N) \quad (X - x) \frac{dx}{ds} + (Y - y) \frac{dy}{ds} + (Z - z) \frac{dz}{ds} = 0,$$

where X , Y , and Z denote the running coordinates. The intersection of this normal plane with the consecutive normal plane is given by this equation together with the following one:

$$(I) \quad (X - x) \frac{d\left(\frac{dx}{ds}\right)}{ds} + (Y - y) \frac{d\left(\frac{dy}{ds}\right)}{ds} + (Z - z) \frac{d\left(\frac{dz}{ds}\right)}{ds} = 1,$$

since

$$\left(\frac{dx}{ds}\right)^2 + \left(\frac{dy}{ds}\right)^2 + \left(\frac{dz}{ds}\right)^2 = 1.$$

Now it is easy to see that plane (I) is perpendicular to plane (N), because

$$\frac{dx}{ds} d\left(\frac{dx}{ds}\right) + \frac{dy}{ds} d\left(\frac{dy}{ds}\right) + \frac{dz}{ds} d\left(\frac{dz}{ds}\right) = 0;$$

hence the perpendicular drawn from (x, y, z) to the intersection of planes (N) and (I) is simply the perpendicular drawn from the same point to plane (I). The radius of curvature is therefore the perpendicular drawn from (x, y, z) to plane (I). This observation yields, very simply, the known theorems concerning the radii of curvature of curves in space.

ANALYSIS

OF A

MEMOIR ON THE ALGEBRAIC SOLUTION OF EQUATIONS^(*).

Equations are called nonprimitive when, being, for example, of degree mn , they decompose into m factors of degree n by means of a single equation of degree m . These are M. Gauss's equations. Primitive equations are those that admit no such simplification. With regard to primitive equations, I have obtained the following results:

1° For an equation of prime degree to be solvable by radicals, it is necessary and sufficient that, once any two of its roots are known, the others can be deduced rationally from them.

2° For a primitive equation of degree m to be solvable by radicals, it is necessary that $m = p^\nu$, where p is a prime number.

3° Apart from the cases mentioned below, for a primitive equation of degree p^ν to be solvable by radicals, it is necessary that, once any two of its roots are known, the others can be deduced rationally from them.

The following very special cases escape the preceding rule:

1° The case $m = p^\nu = 9, = 25$;

2° The case $m = p^\nu = 4$, and generally the case in which, with a^α a divisor of $\frac{p^\nu - 1}{p - 1}$, a is prime and one has

$$\frac{p^\nu - 1}{a^\alpha(p - 1)} \nu = p \pmod{a^\alpha}.$$

These cases nevertheless depart very little from the general rule.

When $m = 9, = 25$, the equation must be of the type of those that determine the trisection and quintisection of elliptic functions.

^(*)M. Férussac's *Bulletin des Sciences mathématiques*, vol. XIII, p. 271 (1830, April issue). (J. LIOUVILLE.)

In the second case it will always be necessary that, once two of the roots are known, the others can be deduced from them, at least by means of as many radicals of degree p as there are divisors a^α of $\frac{p^\nu - 1}{p - 1}$ that satisfy

$$\frac{p^\nu - 1}{a^\alpha(p - 1)} \nu = p \pmod{a^\alpha}, \quad a \text{ prime.}$$

All these propositions were deduced from the theory of permutations.

Here are further results that follow from my theory.

1° Let k be the modulus of an elliptic function and p a given prime number > 3 . For the equation of degree $p + 1$ that gives the various moduli of the functions transformed with respect to the number p to be solvable by radicals, *one of two things is necessary*: either one of the roots must be known rationally, or all the roots must be rational functions of one another. It is of course understood that only particular values of the modulus k are at issue here. It is clear that this does not hold in general. This rule does not hold for $p = 5$.

2° It is remarkable that the general modular equation of the sixth degree corresponding to the number 5 can be reduced to one of the fifth degree of which it is the reduced equation. By contrast, in higher degrees the modular equations cannot be reduced in degree^(*).

^(*)This assertion is not entirely accurate, as Galois himself warns in his Letter to M. Auguste Chevalier, which appears below. Speaking generally of the article reproduced here, he says: "The condition I gave in the *Bulletin de Férussac* for solvability by radicals is too restrictive; there are few exceptions, but there are some." As for modular equations in particular, he declares that the reduction from degree $p + 1$ to degree p is possible not only for $p = 5$, but also for $p = 7$ and $p = 11$; he nevertheless maintains its impossibility for $p > 11$. (J. LIOUVILLE.)

NOTE

ON THE

SOLUTION OF NUMERICAL EQUATIONS (*).

M. Legendre was the first to observe that, when an algebraic equation was put in the form

$$\varphi x = x,$$

where φx is a function of x that increases continually with x , it was easy to find the root of this equation immediately below a given number a if $\varphi a < a$, and the root immediately above a if $\varphi a > a$.

To prove this, construct the curve $y = \varphi x$ and the line $y = x$. Take an abscissa equal to a , and suppose, to fix ideas, that $\varphi a > a$. I say that it will be easy to obtain the root immediately above a . Indeed, the roots of the equation $\varphi x = x$ are simply the abscissas of the points where the line and the curve intersect, and it is clear that one will approach the nearest point of intersection by replacing the abscissa a with the abscissa φa . A still closer value will be obtained by taking $\varphi \varphi a$, then $\varphi \varphi \varphi a$, and so on.

Let $Fx = 0$ be a given equation of degree n , and let $Fx = X - Y$, where X and Y contain only positive terms. Legendre successively puts the equation into these two forms:

$$x = \varphi x = \sqrt[n]{\frac{X}{\left(\frac{Y}{x^n}\right)}}, \quad x = \psi x = \sqrt[n]{\frac{X}{\left(\frac{x^n}{Y}\right)}};$$

the two functions φx and ψx are always, as one sees, one greater and the other less than x . Thus, with the aid of these two functions, one can obtain the two roots of the equation nearest to a given number a , one above and the other below it.

(*) M. Férussac's *Bulletin des Sciences mathématiques*, vol. XIII, p. 413 (1830, June issue). (J. LIOUVILLE.)

But this method has the disadvantage of requiring the extraction of an n th root at each operation. Here are two more convenient forms. Let us seek a number k such that the function

$$x + \frac{Fx}{kx^n}$$

increases with x when $x > 1$. (It is enough, in fact, to know how to find the roots of an equation that are greater than unity.)

For the proposed condition we shall have

$$1 + \frac{d \frac{X - Y}{kx^n}}{dx} > 0, \quad \text{or} \quad 1 - \frac{nX - xX'}{kx^{n+1}} + \frac{nY - xY'}{kx^{n+1}} > 0;$$

but identically we have

$$nX - xX' > 0, \quad nY - xY' > 0;$$

it is therefore enough to set

$$\frac{nX - xX'}{kx^{n+1}} < 1 \quad \text{for} \quad x > 1,$$

and for this it is enough to take as k the value of the function $nX - xX'$ corresponding to $x = 1$.

One similarly finds a number h such that the function

$$x - \frac{Fx}{hx^n}$$

will increase with x when $x > 1$, by replacing Y with X .

Thus the given equation can be put into either of the forms

$$x = x + \frac{Fx}{kx^n}, \quad x = x - \frac{Fx}{hx^n},$$

both of which are rational and provide an easy method of solution.

ON THE THEORY OF NUMBERS^(*).

When one agrees to regard as zero all quantities that, in algebraic calculations, are found multiplied by a given prime number p , and, under this convention, seeks the solutions of an algebraic equation $Fx = 0$, which M. Gauss denotes by $Fx \equiv 0$, it is customary to consider only the integral solutions of such questions. Having been led by particular investigations to consider incommensurable solutions, I have obtained some results that I believe to be new.

Let there be such an equation or congruence, $Fx = 0$, with modulus p . Suppose first, for greater simplicity, that the congruence in question admits no commensurable factor; that is, that one cannot find three functions φx , ψx , and χx such that

$$\varphi x \cdot \psi x = Fx + p \cdot \chi x.$$

In this case the congruence will therefore admit no integral root, nor even any incommensurable root of lower degree. The roots of this congruence must therefore be regarded as a kind of imaginary symbol, since they do not satisfy questions concerning the integers; the use of these symbols in calculation will often be as useful as that of the imaginary $\sqrt{-1}$ in ordinary analysis.

We shall be concerned with classifying these imaginaries and reducing them to the smallest possible number.

Call i one of the roots of the congruence $Fx = 0$, which we suppose to have degree ν .

Consider the general expression

$$a + a_1 i + a_2 i^2 + \cdots + a_{\nu-1} i^{\nu-1}, \tag{A}$$

^(*)M. Férussac's *Bulletin des Sciences mathématiques*, vol. XIII, p. 438 (1830, June issue), with the following note: "This Memoir forms part of M. Galois's research on the theory of permutations and algebraic equations." (J. LIOUVILLE.)

where $a, a_1, a_2, \dots, a_{\nu-1}$ represent integers. When these numbers are given all possible values, expression (A) takes p^ν values, which, as I shall show, possess the same properties as the natural numbers in the *theory of power residues*.

Of the expressions (A), take only the $p^\nu - 1$ values for which $a, a_1, a_2, \dots, a_{\nu-1}$ are not all zero; let α be one of these expressions.

If α is raised successively to the second, third, $\dots$, powers, one obtains a sequence of quantities of the same form [because every function of i can be reduced to degree $\nu - 1$]. Thus one must have $\alpha^n = 1$ for some number n ; let n be the least number for which $\alpha^n = 1$. There will be a set of n mutually distinct expressions,

$$1, \alpha, \alpha^2, \alpha^3, \dots, \alpha^{n-1}.$$

Multiply these n quantities by another expression β of the same form. We obtain another new group of quantities, all different from the first and from one another. If the quantities (A) have not been exhausted, multiply the powers of α again by a new expression γ , and so on. One thus sees that the number n must divide the total number of quantities (A). Since this number is $p^\nu - 1$, one sees that n divides $p^\nu - 1$. It follows further that

$$\alpha^{p^\nu-1} = 1, \quad \text{or} \quad \alpha^{p^\nu} = \alpha.$$

Next one proves, as in the theory of numbers, that there are primitive roots α for which precisely $p^\nu - 1 = n$, and which consequently reproduce, on being raised to powers, the entire sequence of the other roots.

And any one of these primitive roots will depend only on a congruence of degree ν , an *irreducible* congruence; otherwise the equation in i would not be irreducible either, because the roots of the congruence in i are all powers of the primitive root.

We see here the remarkable consequence that all algebraic quantities that can arise in the theory are roots of equations of the form

$$x^{p^\nu} = x.$$

Stated algebraically, this proposition is as follows: given a function Fx and a prime number p , one can write

$$fx \cdot Fx = x^{p^\nu} - x + p\varphi x,$$

where fx and φx are integral functions, whenever the congruence $Fx = 0 \pmod{p}$ is irreducible.

If one wishes to obtain all the roots of such a congruence from a single one, it is enough to observe that, in general,

$$(Fx)^{p^n} = F(x^{p^n})$$

and consequently, if one root is x , the others will be

$$x^p, x^{p^2}, \dots, x^{p^{\nu-1}(1)}.$$

It remains to show that, conversely to what we have just said, all the roots of the equation or congruence $x^{p^\nu} = x$ depend on a single congruence of degree ν .

Indeed, let i be a root of an irreducible congruence such that all the roots of the congruence $x^{p^\nu} = x$ are rational functions of i (it is clear that this property holds here as it does for ordinary equations) ⁽²⁾.

⁽¹⁾From the fact that the roots of the irreducible congruence of degree ν

$$Fx = 0$$

are expressed by the sequence

$$x, x^p, x^{p^2}, \dots, x^{p^{\nu-1}},$$

it would be wrong to conclude that these roots are always quantities expressible by radicals. Here is an example to the contrary:

The irreducible congruence

$$x^2 + x + 1 = 0 \pmod{2}$$

gives

$$x = \frac{-1 + \sqrt{-3}}{2},$$

which reduces to

$$\frac{0}{0} \pmod{2},$$

a formula that tells us nothing.

⁽²⁾The general proposition in question can be stated as follows: given an algebraic equation, one can find a rational function θ of all its roots such that, conversely, each root is expressed rationally in terms of θ . This theorem was known to Abel, as may be seen from the first Part of the Memoir left by that celebrated geometer on elliptic functions.

It is first evident that the degree μ of the congruence in i cannot be less than ν , for otherwise the congruence

$$x^{p^\nu-1} - 1 = 0 \quad (\nu)$$

would have all its roots in common with the congruence

$$x^{p^\mu-1} - 1 = 0,$$

which is absurd, since congruence (ν) has no equal roots, as one sees by taking the derivative of its left-hand side. I now say that μ cannot be greater than ν either.

Indeed, if that were so, all the roots of the congruence

$$x^{p^\mu} = x$$

would have to depend rationally on those of the congruence

$$x^{p^\nu} = x.$$

But it is easy to see that, if

$$i^{p^\nu} = i,$$

every rational function $h = fi$ will also give

$$(fi)^{p^\nu} = f(i^{p^\nu}) = fi, \quad \text{whence} \quad h^{p^\nu} = h.$$

Thus all the roots of the congruence $x^{p^\mu} = x$ would be common to it and to the equation $x^{p^\nu} = x$, which is absurd.

We now know, therefore, that all the roots of the equation or congruence $x^{p^\nu} = x$ necessarily depend on a *single irreducible* congruence of degree ν .

Now, to obtain this irreducible congruence on which the roots of the congruence $x^{p^\nu} = x$ depend, the most general method will be first to free this congruence of all factors it may have in common with congruences of lower degree and of the form

$$x^{p^\mu} = x.$$

One will thereby obtain a congruence that must split into irreducible congruences of degree ν . And since one knows how to express all the roots of each of these irreducible congruences in terms of a single one,

it will be easy to obtain them all by M. Gauss's method.

Most often, however, it will be easy by trial to find an irreducible congruence of a given degree ν , and all the others should be deduced from it.

As an example, let $p = 7$, $\nu = 3$. Let us seek the roots of the congruence

$$x^{7^3} = x \pmod{7}. \quad (1)$$

I observe that the congruence

$$i^3 = 2 \pmod{7} \quad (2)$$

is irreducible and of degree 3, and all the roots of congruence (1) therefore depend rationally on those of congruence (2), so that all the roots of (1) are of the form

$$a + a_1 i + a_2 i^2 \quad \text{or} \quad a + a_1 \sqrt[3]{2} + a_2 \sqrt[3]{4}. \quad (3)$$

We must now find a primitive root; that is, a form of expression (3) which, when raised to all powers, gives all the roots of the congruence

$$x^{7^3-1} = 1, \quad \text{namely} \quad x^{2^1 \cdot 3^2 \cdot 19} = 1 \pmod{7},$$

and for this we need only obtain a primitive root of each congruence

$$x^2 = 1, \quad x^{3^2} = 1, \quad x^{19} = 1.$$

The primitive root of the first is -1 ; those of $x^{3^2} - 1 = 0$ are given by the equations

$$x^3 = 2, \quad x^3 = 4,$$

so that i is a primitive root of $x^{3^2} = 1$.

It remains only to find a root of $x^{19} - 1 = 0$, or rather of

$$\frac{x^{19} - 1}{x - 1} = 0,$$

and for this let us try whether the question can be satisfied

by simply setting $x = a + a_1i$ instead of $a + a_1i + a_2i^2$. We must have

$$(a + a_1i)^{19} = 1,$$

which, on expanding by Newton's formula and reducing the powers of a , a_1 , and i by the formulas

$$a^{m(p-1)} = 1, \quad a_1^{m(p-1)} = 1, \quad i^3 = 2,$$

reduces to

$$3[a - a^4a_1^3 + (a^5a_1^2 + a^2a_1^5)i^2] = 1,$$

whence, on separating,

$$3a - 3a^4a_1^3 = 1, \quad a^5a_1^2 + a^2a_1^5 = 0.$$

These last two equations are satisfied by setting $a = -1$, $a_1 = 1$. Thus

$$-1 + i$$

is a primitive root of $x^{19} = 1$. Above we found the values -1 and i as primitive roots of $x^2 = 1$ and $x^{3^2} = 1$; it remains only to multiply together the three quantities

$$-1, \quad i, \quad -1 + i,$$

and the product $i - i^2$ will be a primitive root of the congruence

$$x^{7^3-1} = 1.$$

Thus here the expression $i - i^2$ has the property that, by raising it to all powers, one obtains $7^3 - 1$ distinct expressions of the form

$$a + a_1i + a_2i^2.$$

If we wish to obtain the congruence of least degree on which our primitive root depends, we must eliminate i between the two equations

$$i^3 = 2, \quad \alpha = i - i^2.$$

We thereby obtain

$$\alpha^3 - \alpha + 2 = 0.$$

It will be convenient to take as a basis for the imaginaries, and to represent by i , a root of this equation, so that

$$i^3 - i + 2 = 0, \tag{i}$$

and all the imaginaries of the form

$$a + a_1 i + a_2 i^2,$$

will be obtained by raising i to all powers and reducing by equation (i).

The principal advantage of the new theory just presented is that it restores to congruences the property (so useful for ordinary equations) of admitting exactly as many roots as their degree.

The method for obtaining all these roots will be very simple. First, one can always prepare the given congruence $Fx = 0$ so that it no longer has equal roots, or, in other words, so that it no longer has a common factor with $F'x = 0$, and the means of doing this is evidently the same as for ordinary equations.

Next, to obtain the integral solutions, it will be enough, as M. Libri appears to have been the first to observe, to seek the greatest common factor of $Fx = 0$ and $x^{p-1} = 1$.

If one now wishes to obtain the imaginary solutions of the second degree, one seeks the greatest common factor of $Fx = 0$ and $x^{p^2-1} = 1$; in general, the solutions of order ν will be given by the greatest common divisor of $Fx = 0$ and $x^{p^\nu-1} = 1$.

It is above all in the theory of permutations, where one constantly needs to vary the form of the indices, that consideration of the imaginary roots of congruences appears indispensable. It provides a simple and easy means of recognizing when a primitive equation is solvable by radicals, as I shall try to indicate briefly.

Let $fx = 0$ be an algebraic equation of degree p^ν ; suppose that the p^ν roots are denoted by x_k , with the index k taking the p^ν values determined by the congruence $k^{p^\nu} = k \pmod{p}$.

Take any rational function V of the p^ν roots x_k . Transform this function by replacing the index k everywhere with the index $(ak + b)^{p^r}$, where a , b , and r are arbitrary constants satisfying $a^{p^\nu-1} = 1$, $b^{p^\nu} = b \pmod{p}$, and r is an integer.

By giving the constants a , b , and r all their possible values, one obtains in all $p^\nu(p^\nu - 1)\nu$ ways of

permuting the roots among themselves by substitutions of the form $[x_k, x_{(ak+b)^{p^r}}]$, and under these substitutions the function V will in general take $p^\nu(p^\nu - 1)\nu$ different forms.

Now suppose that the proposed equation $fx = 0$ is such that every function of the roots invariant under the $p^\nu(p^\nu - 1)\nu$ permutations just constructed thereby has a rational numerical value.

It follows that, under these circumstances, the equation $fx = 0$ will be solvable by radicals; to reach this conclusion, it is enough to observe that the value substituted for k in each index can be written in the three forms

$$(ak + b)^{p^r} = [a(k + b^1)]^{p^r} = a'k^{p^r} + b'' = a'(k + b')^{p^r}.$$

Those accustomed to the theory of equations will see this without difficulty.

This observation would be of little importance had I not succeeded in proving that, conversely, a primitive equation cannot be solvable by radicals unless it satisfies the conditions just stated. (I except equations of the ninth and twenty-fifth degrees.)

Thus, for each number of the form p^ν , one can form a group of permutations such that every function of the roots invariant under these permutations must have a rational value whenever the equation of degree p^ν is primitive and solvable by radicals.

Moreover, only equations of such a degree p^ν are both primitive and solvable by radicals.

The general theorem just stated makes precise and develops the conditions I gave in the April issue of the *Bulletin*. It gives a means of forming a function of the roots whose value will be rational whenever the primitive equation of degree p^ν is solvable by radicals, and consequently leads to criteria for the solvability of these equations by calculations which, if not practicable, are at least possible in theory.

It should be noted that, when $\nu = 1$, the various values of k are nothing other than the sequence of integers. The

substitutions of the form (x_k, x_{ak+b}) will number $p(p-1)$.

The function that, in the case of equations solvable by radicals, must have a rational value will in general depend on an equation of degree $1 \cdot 2 \cdot 3 \cdots (p-2)$, to which the rational-root method must consequently be applied.

II. — POSTHUMOUS WORKS.

LETTER TO AUGUSTE CHEVALIER^(*).

My dear friend,

I have done several new things in analysis.

Some concern the theory of equations; others, integral functions.

In the theory of equations, I investigated the cases in which equations are solvable by radicals. This gave me occasion to study the theory more deeply and to describe all possible transformations of an equation, even when it is not solvable by radicals.

Three Memoirs can be made from all this.

The first is written, and, despite what Poisson said of it, I stand by it, with the corrections I have made.

The second contains some rather curious applications of the theory of equations. Here is a summary of the most important things:

1° From Propositions II and III of the first Memoir, one sees a great difference between adjoining to an equation one root of an auxiliary equation and adjoining all its roots.

In both cases, upon adjunction the group of the equation splits into groups such that one passes from one to another by the same substitution; but the condition that these groups have the same substitutions certainly holds only in the second case. This is called *proper decomposition*.

In other words, when a group G contains another group H ,

^(*)Written on the eve of the author's death. (Published in 1832 in the *Revue encyclopédique*, September issue, page 568.) (J. LIOUVILLE.)

the group G can be split into groups, each obtained by applying the same substitution to the permutations of H , so that

$$G = H + HS + HS' + \dots$$

And it can also be divided into groups that all have the same substitutions, so that

$$G = H + TH + T'H + \dots$$

These two kinds of decomposition do not ordinarily coincide. When they do coincide, the decomposition is called *proper*.

It is easy to see that, when the group of an equation admits no proper decomposition, no matter how the equation is transformed, the groups of the transformed equations will always have the same number of permutations.

By contrast, when the group of an equation admits a proper decomposition, so that it splits into M groups of N permutations, one can solve the given equation by means of two equations: one will have a group of M permutations, the other one of N permutations.

Thus, when one has exhausted on the group of an equation all possible proper decompositions of that group, one arrives at groups that can be transformed but whose permutations will always remain equal in number.

If these groups each have a prime number of permutations, the equation will be solvable by radicals; otherwise, it will not.

The smallest number of permutations that an indecomposable group can have, when this number is not prime, is 5.4.3.

2° The simplest decompositions are those obtained by M. Gauss's method.

Since these decompositions are evident even in the present form of the group of the equation, it is unnecessary to dwell long on this subject.

What decompositions can be carried out on an equation that is not simplified by M. Gauss's method?

I have called *primitive* the equations that cannot be simplified by M. Gauss's method; not that these equations are

actually indecomposable, since they may even be solvable by radicals.

As a lemma for the theory of primitive equations solvable by radicals, I published in June 1830, in the *Bulletin de Férussac*, an analysis of the imaginaries in the theory of numbers.

In the attached material ^(*) one will find the proof of the following theorems:

- 1° For a primitive equation to be solvable by radicals, it must have degree p^ν , where p is prime.
- 2° All the permutations of such an equation are of the form

$$x_{k,l,m,\dots} \mid x_{ak+bl+cm+\dots+h}, a'k+b'l+c'm+\dots+h', a''k+\dots,\dots,$$

where $k, l, m, \dots$ are ν indices which, each taking p values, designate all the roots. The indices are taken modulo p ; that is, the root will be the same when a multiple of p is added to one of the indices.

The group obtained by performing all substitutions of this linear form contains in all

$$p^\nu(p^\nu - 1)(p^\nu - p) \dots (p^\nu - p^{\nu-1}) \text{ permutations.}$$

It is far from being the case that, in this generality, the corresponding equations are solvable by radicals.

The condition I gave in the *Bulletin de Férussac* for an equation to be solvable by radicals is too restrictive; there are few exceptions, but there are some.

The last application of the theory of equations concerns the modular equations of elliptic functions.

It is known that the group of the equation whose roots are the sines of the amplitudes of the $p^2 - 1$ divisions of a period is

$$x_{k,l}, \quad x_{ak+bl, ck+dl};$$

consequently, the corresponding modular equation will have the group

$$x_{\frac{k}{l}}, \quad x_{\frac{ak+bl}{ck+dl}},$$

^(*)Galois is speaking of the hitherto unpublished manuscripts that we are publishing.

in which $\frac{k}{l}$ can take the $p + 1$ values

$$\infty, 0, 1, 2, \dots, p - 1.$$

Thus, agreeing that k may be infinite, one can simply write

$$x_k, \quad x_{\frac{ak+b}{ck+d}}.$$

By giving $a, b, c,$ and d all possible values, one obtains

$$(p + 1)p(p - 1) \text{ permutations.}$$

Now this group decomposes *properly* into two groups whose substitutions are

$$x_k, \quad x_{\frac{ak+b}{ck+d}},$$

where $ad - bc$ is a quadratic residue of p .

The group thus simplified has

$$(p + 1)p^{\frac{p-1}{2}} \text{ permutations.}$$

But it is easy to see that it is no longer properly decomposable unless $p = 2$ or $p = 3$.

Thus, however one transforms the equation, its group will always have the same number of permutations.

But it is interesting to know whether the degree can be lowered.

First, it cannot be lowered below p , since an equation of degree less than p cannot have p as a factor in the number of permutations of its group.

Let us therefore see whether the equation of degree $p + 1$, whose roots x_k are designated by giving k all values including infinity, and whose group has substitutions

$$x_k, \quad x_{\frac{ak+b}{ck+d}},$$

with $ad - bc$ a square, can be lowered to degree p .

For this, the group must decompose (improperly, of course) into p groups of $(p + 1)^{\frac{p-1}{2}}$ permutations each.

Let 0 and ∞ be two associated letters in one of these groups. The substitutions that do not interchange 0 and ∞ will be of the form

$$x_k, \quad x_{m^2 k}.$$

Thus, if M is the letter associated with 1, the letter associated with m^2 will be $m^2 M$. When M is a square, we shall therefore have $M^2 = 1$. But this simplification can occur only for $p = 5$.

For $p = 7$ one finds a group of $(p + 1)\frac{p-1}{2}$ permutations in which

$$\infty, 1, 2, 4$$

have respectively as their associated letters

$$0, 3, 6, 5.$$

This group has substitutions of the form

$$x_k, \quad x_{a \frac{k-b}{k-c}},$$

where b is the letter associated with c , and a is a letter that is a residue or a nonresidue at the same time as c .

For $p = 11$, the same substitutions hold with the same notation,

$$\infty, 1, 3, 5, 5, 9$$

having respectively as their associated letters

$$0, 2, 6, 8, 10, 7.$$

Thus, for $p = 5, 7$, and 11 , the modular equation is lowered to degree p .

Strictly speaking, this equation is not possible in the higher cases.

The third Memoir concerns integrals.

It is known that a sum of terms of the same elliptic function always reduces to a single term, together with algebraic or logarithmic quantities.

There are no other functions for which this property holds.

But entirely similar properties take its place for all integrals of algebraic functions.

One treats simultaneously all integrals whose differential is a function of the variable and of the same irrational function of the variable, whether that irrational function is or is not a radical, and whether it can or cannot be expressed by radicals.

One finds that the number of distinct periods of the most general integral relative to a given irrational function is always even.

Let this number be $2n$; then one has the following theorem:

Any sum of terms reduces to n terms, together with algebraic and logarithmic quantities.

Functions of the first kind are those for which the algebraic and logarithmic part is zero.

There are n distinct such functions.

Functions of the second kind are those for which the complementary part is purely algebraic.

There are n distinct such functions.

One may suppose that the differentials of the other functions are never infinite except once, at $x = a$, and, moreover, that their complementary part reduces to a single logarithm, $\log P$, where P is an algebraic quantity. Denoting these functions by $\Pi(x, a)$, one has the theorem

$$\Pi(x, a) - \Pi(a, x) = \sum \varphi a \cdot \psi x,$$

where φa and ψx are functions of the first and second kinds.

Calling $\Pi(a)$ and ψ the periods of $\Pi(x, a)$ and ψx corresponding to the same circuit of x , one deduces

$$\Pi(a) = \sum \psi \times \varphi a.$$

Thus the periods of functions of the third kind are always expressed in terms of functions of the first and second kinds.

One may also deduce theorems analogous to Legendre's theorem

$$FE' + EF' - FF' = \frac{\pi}{2}.$$

The reduction of functions of the third kind to definite integrals, which is M. Jacobi's finest discovery, is not practicable outside the case of elliptic functions.

The multiplication of integral functions by an integer

is always possible, like addition, by means of an equation of degree n whose roots are the values to be substituted in the integral to obtain the reduced terms.

The equation that gives the division of the periods into p equal parts has degree $p^{2n} - 1$. Its group has in all

$$(p^{2n} - 1)(p^{2n} - p) \dots (p^{2n} - p^{2n-1}) \text{ permutations.}$$

The equation that gives the division of a sum of n terms into p equal parts has degree p^{2n} . It is solvable by radicals.

On transformation. — First, by following arguments analogous to those Abel recorded in his last Memoir, one can prove that if, in the same relation between integrals, one has the two functions

$$\int \Phi(x, X) dx, \quad \int \Psi(y, Y) dy,$$

the latter integral having $2n$ periods, one may suppose that y and Y are expressed, by means of a single equation of degree n , as functions of x and X .

Accordingly, one may suppose that transformations always take place between only two integrals, since, on taking any rational function of y and Y , one will evidently have

$$\sum \int f(y, Y) dy = \int F(x, X) dx + \text{an alg. and log. quantity.}$$

There would be evident reductions in this equation if the integrals on the two sides did not both have the same number of periods.

Thus we need compare only integrals that both have the same number of periods.

It will be proved that the least degree of irrationality of two such integrals cannot be greater for one than for the other.

It will then be shown that one can always transform a given integral into another in which one period of the first is divided by the prime number p , while the other $2n - 1$ periods remain the same.

It will therefore remain only to compare integrals whose periods are the same on both sides and such, consequently, that n terms of one are expressed, without any equation other than a single one of degree n , in terms of those of the other, and conversely. Here we know nothing.

You know, my dear Auguste, that these subjects are not the only ones I have explored. For some time my principal meditations have been directed toward applying the theory of ambiguity to transcendental analysis. The question was to see *a priori*, in a relation among transcendental quantities or functions, what interchanges could be made, what quantities could be substituted for the given quantities, without the relation ceasing to hold. This immediately reveals the impossibility of many expressions one might seek. But I have no time, and my ideas are not yet well developed in this immense field.

You will have this Letter printed in the *Revue encyclopédique*.

I have often ventured in my life to put forward propositions of which I was not certain; but everything I have written here has been in my mind for nearly a year, and it is too much in my interest not to be mistaken for anyone to suspect me of stating theorems of which I did not have a complete proof.

You will publicly ask Jacobi or Gauss to give their opinion, not on the truth, but on the importance of the theorems.

After that, I hope there will be people who find it profitable to decipher all this mess.

I embrace you warmly.

E. GALOIS.

29 May 1832.

MEMOIR

ON THE

CONDITIONS FOR THE SOLVABILITY OF EQUATIONS BY RADICALS^(*).

The accompanying Memoir⁽²⁾ is extracted from a work that I had the honor of presenting to the Academy a year ago. Since that work was not understood and the propositions it contains were called into question, I have had to content myself with giving, in synthetic form, the general principles and a *single* application of my theory. I beg my judges at least to read these few pages attentively.

Here one will find a general *condition satisfied by every equation solvable by radicals*, and which, conversely, guarantees solvability. It is applied only to

(*) This Memoir and the one following it were found among Galois's papers and were first published in 1846 by Liouville, who prefaced them with the following note:

"When they inserted in their periodical the letter just read, the editors of the *Revue encyclopédique* announced that they would soon publish the manuscripts left by Galois. But this promise was not kept. M. Auguste Chevalier had nevertheless prepared the work. He has given it to us, and in the sheets that follow one will find:

"1° A complete Memoir on the conditions for the solvability of equations by radicals, with the application to equations of prime degree;

"2° A fragment of a second Memoir in which Galois treats the general theory of equations that he calls *primitive*.

"We have retained most of the notes that M. Auguste Chevalier appended to the Memoirs just mentioned. These notes are all marked with the initials A. Ch. The unsigned notes are by Galois himself.

"We shall complete this publication with a few other pieces extracted from Galois's papers which, though not of great importance, may nevertheless still be read with interest by geometers."

The extracts mentioned by Liouville in the last sentence of this note were never published.

(2) I thought it appropriate to place at the head of this Memoir the preface about to be read, although I found it crossed out in the manuscript.

(A. Ch.)
E. G.

equations whose degree is a prime number. Here is the theorem yielded by our analysis:

For an equation of prime degree having no commensurable divisors to be solvable by radicals, it is necessary and sufficient that all its roots be rational functions of any two of them.

The other applications of the theory each constitute a separate special theory. They also require the use of number theory and of a particular algorithm; we reserve them for another occasion. They concern in part the modular equations of the theory of elliptic functions, which we prove cannot be solved by radicals.

16 January 1831.

E. GALOIS.

PRINCIPLES.

I shall begin by establishing a few definitions and a sequence of lemmas, all of which are known.

Definitions. — An equation is called *reducible* when it has rational divisors, and *irreducible* otherwise.

Here it is necessary to explain what is meant by the word *rational*, since it will occur often.

When the equation has *all* its coefficients numerical and rational, this means simply that the equation can be decomposed into factors whose coefficients are numerical and rational.

But when the coefficients of an equation are *not all* numerical and rational, a rational divisor must then be understood to mean a divisor whose coefficients can be expressed as rational functions of the coefficients of the given equation.

More generally, one may agree to regard as rational every rational function of a certain number of specified quantities assumed to be known *a priori*. For example, one may

choose a particular root of an integer and regard every rational function of that radical as rational.

When we agree in this way to regard certain quantities as known, we shall say that we *adjoin* them to the equation to be solved. We shall say that these quantities are *adjoined* to the equation.

This being established, we shall call *rational* every quantity that can be expressed as a rational function of the coefficients of the equation and of a certain number of quantities *adjoined* to the equation and chosen arbitrarily.

When we use auxiliary equations, they will be rational if their coefficients are rational in our sense.

It is clear, moreover, that the properties and difficulties of an equation may be entirely different according to the quantities adjoined to it. For example, adjoining one quantity may make an irreducible equation reducible.

Thus, when one adjoins to the equation

$$\frac{x^n - 1}{x - 1} = 0.$$

where n is prime, a root of one of M. Gauss's auxiliary equations, the equation decomposes into factors and consequently becomes reducible.

Substitutions are the passage from one permutation to another.

The permutation from which one starts in order to specify the substitutions is entirely arbitrary when functions are concerned, for there is no reason that, in a function of several letters, one letter should occupy one position rather than another.

Nevertheless, because it is difficult to form the idea of a substitution without also forming that of a permutation, we shall frequently use permutations in our terminology and shall regard substitutions only as the passage from one permutation to another.

When we wish to group substitutions, we shall make them all arise from the same permutation.

Since the original arrangement of the letters has no effect in the questions concerning the groups that we shall consider, the substitutions must be the same whatever

the permutation from which one starts. Thus, if such a group contains the substitutions S and T , it certainly contains the substitution ST .

These are the definitions we thought it necessary to recall.

LEMMA I. — *An irreducible equation cannot have a root in common with a rational equation without dividing it.*

For the greatest common divisor of the irreducible equation and the other equation will itself be rational; hence, etc.

LEMMA II. — *Given any equation without repeated roots, whose roots are $a, b, c, \dots$, one can always form a function V of the roots such that no two values obtained by permuting the roots in that function in every possible way are equal.*

For example, one may take

$$V = Aa + Bb + Cc + \dots,$$

where A, B, C are suitably chosen integers.

LEMMA III. — *When the function V has been chosen as indicated in the preceding article, it will have the property that all the roots of the given equation can be expressed rationally as functions of V .*

Indeed, let

$$V = \varphi(a, b, c, d, \dots),$$

or

$$V - \varphi(a, b, c, d, \dots) = 0.$$

Multiply together all similar equations obtained by permuting all the letters in these equations while leaving only the first fixed; the following expression results:

$$\begin{aligned} &[V - \varphi(a, b, c, d, \dots)][V - \varphi(a, c, b, d, \dots)] \\ &[V - \varphi(a, b, d, c, \dots)] \dots, \end{aligned}$$

It is symmetric in $b, c, d, \dots$ and can therefore be written as a function of a . We shall thus have an equation of the form

$$F(V, a) = 0.$$

I now say that the value of a can be obtained from this equation. For this it is enough to seek the common solution of this equation and the given equation. That solution is their only common one, for one cannot have, for example,

$$F(V, b) = 0,$$

with this equation having a common factor with the similar equation; otherwise one of the functions $\varphi(a, \dots)$ would equal one of the functions $\varphi(b, \dots)$, contrary to the hypothesis.

It follows that a can be expressed as a rational function of V , and the same is true of the other roots.

This proposition^(*) is cited without proof by Abel in the posthumous Memoir on elliptic functions.

LEMMA IV. — *Suppose that the equation in V has been formed and one of its irreducible factors has been selected, so that V is a root of an irreducible equation. Let $V, V', V'', \dots$ be the roots of this irreducible equation. If $a = f(V)$ is one root of the given equation, then $f(V')$ will likewise be a root of the given equation.*

Indeed, multiplying together all factors of the form $V - \varphi(a, b, c, \dots, d)$ obtained by making all possible permutations of the letters gives a rational equation in V , which is necessarily divisible by the equation in question; hence V' must be obtained by exchanging the letters in the function V . Let $F(V, a) = 0$ be the equation obtained by permuting in V all the letters except the first. We shall then have $F(V', b) = 0$, where b may equal a but is certainly one of the roots of the given equation. Just as the given equation together with $F(V, a) = 0$ yielded $a = f(V)$, so the given equation together with $F(V', b) = 0$ yields $b = f(V')$.

^(*) It is remarkable that this proposition implies that every equation depends on an auxiliary equation whose roots are all rational functions of one another, for the auxiliary equation in V is such an equation.

This observation is, however, merely curious. An equation having this property is not in general easier to solve than any other.

PROPOSITION I.

THEOREM. — *Let an equation be given whose m roots are $a, b, c, \dots$. There will always be a group of permutations of the letters $a, b, c, \dots$ having the following property:*

1° Every function of the roots invariant^() under the substitutions of this group is rationally known;*

2° Conversely, every function of the roots that is rationally determinate is invariant under the substitutions.

(For algebraic equations, this group is nothing other than the set of the $1.2.3 \dots m$ possible permutations of the m letters, since in that case only symmetric functions are rationally determinate.)

(For the equation $\frac{x^n - 1}{x - 1} = 0$, if one sets $a = r$, $b = r^g$, $c = r^{g^2}$, $\dots$, where g is a primitive root, the group of permutations is simply the following:

$abcd \dots k,$
 $bcd \dots ka,$
 $cd \dots kab,$
 $\dots \dots \dots$
 $kabc \dots i;$

in this special case the number of permutations equals the degree of the equation, and the same would hold for equations whose roots are all rational functions of one another.)

Proof. — Whatever equation is given, one can find a rational function V of the roots such that all

(*) Here we call invariant not only a function whose form is invariant under substitutions of the roots among themselves, but also one whose numerical value does not vary under those substitutions. For example, if $Fx = 0$ is an equation, Fx is a function of the roots that varies under no permutation.

When we say that a function is rationally known, we mean that its numerical value can be expressed as a rational function of the coefficients of the equation and of the adjoined quantities.

the roots are rational functions of V . This being established, consider the irreducible equation of which V is a root (Lemmas III and IV). Let $V, V', V'', \dots, V^{(n-1)}$ be the roots of this equation.

Let $\varphi V, \varphi_1 V, \varphi_2 V, \dots, \varphi_{m-1} V$ be the roots of the given equation.

Write the following n permutations of the roots:

$$\begin{array}{cccccc} (V) & \varphi V & \varphi_1 V & \varphi_2 V & \dots & \varphi_{m-1} V, \\ (V') & \varphi V' & \varphi_1 V' & \varphi_2 V' & \dots & \varphi_{m-1} V', \\ (V'') & \varphi V'' & \varphi_1 V'' & \varphi_2 V'' & \dots & \varphi_{m-1} V'', \\ \dots & \dots & \dots & \dots & \dots & \dots, \\ (V^{(n-1)}) & \varphi V^{(n-1)} & \varphi_1 V^{(n-1)} & \varphi_2 V^{(n-1)} & \dots & \varphi_{m-1} V^{(n-1)} : \end{array}$$

I say that this group of permutations has the stated property.

Indeed:

1° Every function F of the roots invariant under the substitutions of this group can be written $F = \psi V$, and we shall have

$$\psi V = \psi V' = \psi V'' = \dots = \psi V^{(n-1)}.$$

The value of F can therefore be determined rationally.

2° Conversely, if a function F is rationally determinate and we set $F = \psi V$, we must have

$$\psi V = \psi V' = \psi V'' = \dots = \psi V^{(n-1)},$$

since the equation in V has no commensurable divisor and V satisfies the equation $F = \psi V$, with F a rational quantity. Thus the function F will necessarily be invariant under the substitutions of the group written above.

Hence this group has the twofold property stated in the proposed theorem. The theorem is therefore proved.

We shall call the group in question the *group of the equation*.

Scholium I. — It is clear that, in the group of permutations considered here, the arrangement of the letters is immaterial; only the substitutions of letters by which one passes from one permutation to another are to be considered.

Thus one may choose a first permutation arbitrarily, provided that all the other permutations are always derived from it by the same substitutions of letters. The new group so formed will plainly have the same properties as the first,

since the preceding theorem concerns only the substitutions that can be made in the functions.

Scholium II. — The substitutions are independent even of the number of roots.

PROPOSITION II.

THEOREM^(*). — *If one adjoins to a given equation a root r of an irreducible auxiliary equation, 1° one of two things will happen: either the group of the equation will remain unchanged, or it will split into p groups belonging respectively to the given equation when each root of the auxiliary equation is adjoined; 2° these groups will have the remarkable property that one passes from one to another by making the same substitution of letters in every permutation of the first.*

1° If, after r is adjoined, the equation in V considered above remains irreducible, it is clear that the group of the equation will remain unchanged. If, on the contrary, it becomes reducible, the equation in V will then decompose into p factors, all of the same degree and of the form

$$f(V, r) \times f(V, r') \times f(V, r'') \times \dots,$$

where $r, r', r'', \dots$ are other values of r . Thus the group of the given equation will also decompose into groups, each containing the same number of permutations, since each value of V corresponds to a permutation. These groups will be respectively

(*) In the statement of the theorem, after the words “the root r of an irreducible auxiliary equation,” Galois first wrote “of prime degree p ,” which he later erased. Likewise, in the proof, instead of “ $r, r', r'', \dots$ being other values of r ,” the original wording read “ $r, r', r'', \dots$ being the various values of r .” Finally, the following note by the author is found in the margin of the manuscript: “There is something to complete in this proof. I do not have time.”

This line was dashed onto the paper very rapidly; that circumstance, together with the words “I do not have time,” makes me think that Galois reread his Memoir to correct it before going to the field. (A. CH.)

those of the given equation when $r, r', r'', \dots$ are adjoined successively.

2° We saw above that all the values of V were rational functions of one another. Accordingly, suppose that, when V is a root of $f(V, r) = 0$, $F(V)$ is another root; it is clear that, likewise, if V' is a root of $f(V, r') = 0$, then $F(V')$ will be another, for we shall have

$$f[F(V), r] = \text{a function divisible by } f(V, r).$$

Therefore (*Lemma I*)

$$f[F(V'), r'] = \text{a function divisible by } f(V', r').$$

This being established, I say that the group relative to r' is obtained by making everywhere in the group relative to r one and the same substitution of letters.

Indeed, if, for example, we have

$$\varphi_\mu F(V) = \varphi_\nu(V),$$

then we shall also have (*Lemma I*)

$$\varphi_\mu F(V') = \varphi_\nu(V').$$

Thus, to pass from the permutation $[F(V)]$ to the permutation $[F(V')]$, one must make the same substitution as is made in passing from the permutation (V) to the permutation (V') .

The theorem is therefore proved.

PROPOSITION III.

THEOREM. — *If all the roots of an auxiliary equation are adjoined to an equation, the groups considered in Theorem II will have in addition the property that the substitutions are the same in each group.*

The proof will be found^(*).

^(*) In the manuscript, the statement of the theorem just read is in the margin and replaces another that Galois had written with its proof under the same title, *Proposition III*. Here is the original text: THEOREM. — If the equa-

tion in r is of the form $r^p = A$, and if the p th roots of unity are among the quantities previously adjoined, the p groups considered in Theorem II will have, in addition, this property that the substitutions of letters by which one passes from one permutation to another in each group are the same for all the groups. Indeed, in this case it is immaterial whether one adjoins this or that value of r to the equation. Consequently, its properties must be the same after this or that value is adjoined. Thus its group must be the same with respect to substitutions (Proposition I, Scholium). Hence, etc.

All this is carefully erased; the new statement bears the date 1832 and shows, by the way it is written, that the author was in an extreme hurry, which confirms the assertion I made in the preceding Note. (A. CH.)

PROPOSITION IV.

THEOREM. — *If the numerical value of a certain function of the roots is adjoined to an equation, the group of the equation will be reduced so that it contains only those permutations under which that function is invariant.*

Indeed, by Proposition I, every known function must be invariant under the permutations of the group of the equation.

PROPOSITION V.

PROBLEM. — *Under what condition is an equation solvable by simple radicals?*

I shall first observe that, to solve an equation, its group must successively be reduced until it contains only one permutation. For when an equation is solved, every function of its roots is known, even when it is invariant under no permutation.

This being established, let us seek the condition that the group of an equation must satisfy in order that it may be reduced in this way by adjoining radical quantities.

Let us follow the sequence of possible operations in this solution, treating the extraction of each root of prime degree as a distinct operation.

Adjoin to the equation the first radical extracted in the solution. Two cases may occur: either adjoining this radical diminishes the group of permutations of the equation, or, the extraction being merely preparatory, the group remains the same.

In any event, after some *finite* number of root extractions the group must be diminished, for otherwise the equation would not be solvable.

If, upon reaching this point, there were several ways to diminish the group of the given equation by a single root extraction, then, for what we are about to say, one should consider only a radical of the lowest possible degree among all the simple radicals whose individual adjunction diminishes the group of the equation.

Let p therefore be the prime number representing this minimum degree, so that the group of the equation is diminished by extracting a root of degree p .

We may always suppose, at least as far as the group of the equation is concerned, that a p th root of unity, α , is among the quantities previously adjoined to the equation. Since this expression is obtained by extractions of roots of degree less than p , knowing it will not alter the group of the equation in any way.

Consequently, by Theorems II and III, the group of the equation must decompose into p groups having, relative to one another, this twofold property: 1° one passes from one to another by one and the same substitution; 2° all contain the same substitutions.

Conversely, I say that if the group of the equation can be divided into p groups having this twofold property, then by one extraction of a p th root and the adjunction of that p th root the group of the equation can be reduced to one of these partial groups.

Indeed, take a function of the roots that is invariant under every substitution of one partial group and varies under every other substitution. (For this it is enough to choose a function symmetric in the various values taken, under all the

permutations of one partial group, by a function invariant under no substitution.)

Let θ be this function of the roots.

Apply to θ one of the substitutions of the total group that is not common to the partial groups. Let θ_1 be the result. Apply the same substitution to θ_1 and let θ_2 be the result, and so on.

Since p is prime, this sequence cannot stop before the term θ_{p-1} ; after that we shall have

$$\theta_p = \theta_0, \quad \theta_{p+1} = \theta_1,$$

and so on.

This being established, it is clear that the function

$$(\theta + \alpha\theta_1 + \alpha^2\theta_2 + \dots + \alpha^{p-1}\theta_{p-1})^p$$

will be invariant under all permutations of the total group and consequently will thereby be known.

If the p th root of this function is extracted and adjoined to the equation, then, by Proposition IV, the group of the equation will contain no substitutions other than those of the partial groups.

Thus the condition above is necessary and sufficient for the group of an equation to be reducible by a single root extraction.

Adjoin the radical in question to the equation; we can now reason about the new group as about the preceding one, and it must itself decompose in the indicated way, and so on, until a group is reached that contains only one permutation.

Scholium. — This procedure is readily observed in the known solution of general equations of the fourth degree. Indeed, these equations are solved by means of an equation of the third degree, which itself requires the extraction of a square root. In the natural order of ideas, one must therefore begin with this square root. On adjoining this square root to the equation of the fourth degree, the group of the equation, which contained twenty-four substitutions in all, decomposes into two groups containing only twelve. Denoting the roots by a, b, c, d ,

here is one of these groups:

$$\begin{array}{lll} abcd, & acdb, & adbc, \\ badc, & cabd, & dacb, \\ cdab, & dbac, & bcad, \\ dcba, & bdca, & cbda. \end{array}$$

This group now divides in turn into three groups, as indicated in Theorems II and III. Thus, after extracting a single radical of the third degree, there remains simply the group

$$\begin{array}{l} abcd, \\ badc, \\ cdab, \\ dcba : \end{array}$$

this group divides again into two groups:

$$\begin{array}{ll} abcd, & cdab, \\ badc, & dcba. \end{array}$$

Thus, after a single extraction of a square root, there remains

$$\begin{array}{l} abcd, \\ badc; \end{array}$$

which is finally solved by a single extraction of a square root.

One thus obtains either Descartes's solution or Euler's; for although, after solving the auxiliary equation of the third degree, Euler extracts three square roots, it is known that two suffice, since the third follows rationally from them.

We shall now apply this condition to irreducible equations of prime degree.

Application to irreducible equations of prime degree.

PROPOSITION VI.

LEMMA. — *An irreducible equation of prime degree cannot become reducible by adjoining a radical whose index differs from the degree of the equation itself.*

For if $r, r', r'', \dots$ are the various values of the radical and $Fx = 0$ is the given equation, then Fx would have to divide into factors

$$f(x, r) \times f(x, r') \times \dots,$$

all of the same degree, which is impossible unless $f(x, r)$ is of the first degree in x .

Thus an irreducible equation of prime degree cannot become reducible unless its group is reduced to a single permutation.

PROPOSITION VII.

PROBLEM. — *What is the group of an irreducible equation of prime degree n that is solvable by radicals?*

By the preceding proposition, the smallest possible group before the group containing only one permutation will contain n permutations. Now, a group of permutations of a prime number n of letters cannot be reduced to n permutations unless one of these permutations is obtained from another by a cyclic substitution of order n . (See M. Cauchy's Memoir, *Journal de l'École Polytechnique*, 17th issue.) Thus the penultimate group will be

$$(G) \quad \left\{ \begin{array}{cccccccc} x_0, & x_1, & x_2, & x_3, & \dots, & x_{n-3}, & x_{n-2}, & x_{n-1}, \\ x_1, & x_2, & x_3, & x_4, & \dots, & x_{n-2}, & x_{n-1}, & x_0, \\ x_2, & x_3, & \dots, & \dots, & \dots, & x_{n-1}, & x_0, & x_1, \\ \dots, & \dots, & \dots, & \dots, & \dots, & \dots, & \dots, & \dots, \\ x_{n-1}, & x_0, & x_1, & \dots, & \dots, & x_{n-4}, & x_{n-3}, & x_{n-2}. \end{array} \right.$$

where $x_0, x_1, x_2, \dots, x_{n-1}$ are the roots.

The group immediately preceding this one in the order of decompositions must now consist of a certain number of groups all having the same substitutions as this one. I observe that these substitutions may be expressed as follows (set, in general, $x_n = x_0$, $x_{n+1} = x_1$, ...; it is clear that each substitution of group (G) is obtained by replacing x_k everywhere by x_{k+c} , where c is a constant).

Consider any one of the groups similar to group (G). By Theorem II, it must be obtained by making one and the same substitution everywhere in this group; for example, by replacing x_k everywhere in group (G) by $x_{f(k)}$, where f is a certain function.

Since the substitutions of these new groups must be the same as those of group (G), we must have

$$f(k+c) = f(k) + C,$$

where C is independent of k .

Therefore

$$f(k+2c) = f(k) + 2C,$$

$$f(k+mc) = f(k) + mC.$$

If $c = 1$ and $k = 0$, we find

$$f(m) = am + b$$

or

$$f(k) = ak + b,$$

where a and b are constants.

Thus the group immediately preceding group (G) can contain only substitutions such as

$$x_k, \quad x_{ak+b},$$

and consequently will contain no cyclic substitution other than that of group (G).

One reasons about this group as about the preceding one, and it follows that the first group in the order of decompositions—that is, the *actual* group of the equation—can contain only substitutions of the form

$$x_k, \quad x_{ak+b}.$$

Thus, *if an irreducible equation of prime degree is solvable by radicals, the group of that equation can contain only substitutions of the form*

$$x_k, \quad x_{ak+b},$$

where a and b are constants.

Conversely, if this condition holds, I say that the equation will be solvable by radicals. Indeed, consider the functions

$$\begin{aligned} (x_0 + \alpha x_1 + \alpha^2 x_2 + \dots + \alpha^{n-1} x_{n-1})^n &= X_1, \\ (x_0 + \alpha x_a + \alpha^2 x_{2a} + \dots + \alpha^{n-1} x_{(n-1)a})^n &= X_a, \\ (x_0 + \alpha x_{a^2} + \alpha^2 x_{2a^2} + \dots + \alpha^{n-1} x_{(n-1)a^2})^n &= X_{a^2}, \\ &\dots \end{aligned}$$

where α is an n th root of unity and a is a primitive root modulo n .

It is clear that every function invariant under cyclic substitutions of the quantities $X_1, X_a, X_{a^2}, \dots$ will in this case be immediately known. Thus one can find $X_1, X_a, X_{a^2}, \dots$ by M. Gauss's method for binomial equations. Hence, etc.

Therefore, for an irreducible equation of prime degree to be solvable by radicals, it is *necessary* and *sufficient* that every function invariant under the substitutions

$$x_k, \quad x_{ak+b}$$

be rationally known.

Thus the function

$$(X_1 - X)(X_a - X)(X_{a^2} - X) \dots$$

must be known for every X .

It is therefore *necessary* and *sufficient* that the equation giving this function of the roots have a rational value for every X .

If the given equation has all its coefficients rational, the auxiliary equation giving this function will likewise have all its coefficients rational, and it will suffice to determine whether this auxiliary equation of degree $1.2.3\dots(n-2)$ has a rational root, which one knows how to do.

This is the method that should be used in practice. But we shall present the theorem in another form.

PROPOSITION VIII.

THEOREM. — *For an irreducible equation of prime degree to be solvable by radicals, it is necessary and sufficient that, when any two roots are known, all the others follow rationally from them.*

First, it is necessary, for the substitution

$$x_k, \quad x_{ak+b}$$

never leaving two letters in the same place, it is clear that, after two roots are adjoined to the equation, Proposition IV requires its group to reduce to a single permutation.

Second, it is sufficient; for in this case no substitution of the group will leave two letters in the same places. Consequently, the group will contain at most $n(n-1)$ permutations. Thus it will contain only one cyclic substitution (otherwise there would be at least n^2 permutations). Hence every substitution of the group, $x_k, x_{f(k)}$, must satisfy the condition

$$f(k+c) = f(k) + C.$$

Hence, etc.

The theorem is therefore proved.

E. G.

EXAMPLE OF THEOREM VII.

Let $n = 5$; the group will be as follows:

abcde
bcdea
cdeab
deabc
eabcd

acebd
cebda
ebdac
bdace
daceb

aedcb
edcba
dcbae
cbaed
baedc

adbec
dbeca
becad
ecadb
cadbe

ON PRIMITIVE EQUATIONS
 THAT ARE SOLVABLE BY RADICALS (*)
 (Fragment.)

Let us seek, in general, the circumstances in which a primitive equation is solvable by radicals. We can at once establish a general characteristic based on the degree of such equations. It is this: *For a primitive equation to be solvable by radicals, its degree must have the form p^ν , where p is prime.* It follows immediately that when an irreducible equation whose degree has unequal prime factors is to be solved by radicals, this can be done only by the decomposition method due to M. Gauss; otherwise the equation will be insoluble.

To establish the general property just stated for primitive equations solvable by radicals, we may suppose that the equation to be solved is primitive but ceases to be so upon adjoining a single radical. In other words, we may suppose that, with n prime, the group of the equation divides into n conjugate irreducible but nonprimitive groups. For, unless the degree of the equation is prime, such a group will always occur in the sequence of decompositions.

Let N be the degree of the equation, and suppose that after extracting a root of prime degree n , it becomes nonprimitive and divides into Q primitive equations of degree P by means of a single equation of degree Q .

If we call the group of the equation G , this group must divide into n conjugate nonprimitive groups in which the letters are arranged in systems, each consisting of P connected letters. Let us see in how many ways this can be done.

Let H be one of the conjugate nonprimitive groups. It is easy to

(*) See Note 1 on page 33.

see that in this group any two letters chosen at will belong to some system of P connected letters, and belong to only one such system.

For, in the first place, if there were two letters that could not belong to the same system of P connected letters, the group G —which is such that any one of its substitutions transforms all the substitutions of H into one another—would be nonprimitive, contrary to the hypothesis.

Second, if two letters belonged to several different systems, it would follow that the groups corresponding to the various systems of P connected letters were not primitive, again contrary to the hypothesis.

Accordingly, let

$$\begin{array}{cccccc} a_0, & a_1, & a_2, & \dots, & a_{P-1}, \\ b_0, & b_1, & b_2, & \dots, & b_{P-1}, \\ c_0, & c_1, & c_2, & \dots, & c_{P-1} \end{array}$$

be the N letters; suppose that each horizontal row represents a system of connected letters. Let

$$a_0, \quad a_{0,1}, \quad a_{0,2}, \quad \dots, \quad a_{0,P-1}$$

be P connected letters, all situated in the first vertical column. (It is clear that we may arrange this by interchanging the order of the horizontal rows.) Likewise, let

$$a_{1,0}, \quad a_{1,1}, \quad a_{1,2}, \quad a_{1,3}, \quad \dots, \quad a_{1,P-1}$$

be P connected letters, all situated in the second vertical column, so that

$$a_{1,0}, \quad a_{1,1}, \quad a_{1,2}, \quad a_{1,3}, \quad \dots, \quad a_{1,P-1}$$

respectively belong to the same horizontal rows as

$$a_{0,0}, \quad a_{0,1}, \quad a_{0,2}, \quad a_{0,3}, \quad \dots, \quad a_{0,P-1};$$

and similarly let the systems of connected letters be

$$\begin{array}{cccccc} a_{2,0}, & a_{2,1}, & a_{2,2}, & a_{2,3}, & \dots, & a_{2,P-1}, \\ a_{3,0}, & a_{3,1}, & a_{3,2}, & a_{3,3}, & \dots, & a_{3,P-1}, \\ \dots, & \dots, & \dots, & \dots, & \dots, & \dots, \end{array}$$

we thereby obtain P^2 letters in all. If the total number of letters is not exhausted, we introduce a third index, so that

$$a_{m,n,0}, \quad a_{m,n,1}, \quad a_{m,n,2}, \quad a_{m,n,3}, \quad \dots, \quad a_{m,n,P-1}$$

is in general a system of connected letters. We thereby reach the conclusion that $N = P^\mu$, where μ is the number of distinct indices required. The general form of the letters will be

$$a_{k_1, k_1, k_3, \dots, k_\mu},$$

where $k_1, k_2, k_3, \dots, k_\mu$ are indices, each of which may take the P values $0, 1, 2, 3, \dots, P-1$.

The manner in which we proceeded also shows that every substitution in the group H has the form

$$\left[a_{k_1, k_2, k_3 \dots k_\mu}, \quad a_{\phi(k_1), \psi(k_2), \chi(k_3) \dots, \sigma(k_\mu)} \right],$$

because each index corresponds to a system of connected letters.

If P is not prime, we reason about the permutation group of any one of the systems of connected letters as we did about G , replacing each index by a certain number of new indices. We find $P = R^\alpha$, and so on; ultimately $N = p^\nu$, where p is prime.

Primitive equations of degree p^2 .

Let us pause to treat at once the primitive equations of degree p^2 , where p is an odd prime. (The case $p = 2$ has been examined.) If an equation of degree p^2 is solvable by radicals, suppose first that it becomes nonprimitive upon extracting a radical.

Let G , then, be a primitive group on p^2 letters which divides into n nonprimitive groups conjugate to H .

The letters in H must necessarily be arranged

as follows:

$$\begin{array}{cccccc}
a_{0,0}, & a_{0,1}, & a_{0,2}, & a_{0,3}, & \dots, & a_{0,p-1}, \\
a_{1,0}, & a_{1,1}, & a_{1,2}, & a_{1,3}, & \dots, & a_{1,p-1}, \\
a_{2,0}, & a_{2,1}, & a_{2,2}, & a_{2,3}, & \dots, & a_{2,p-1}, \\
\dots, & \dots, & \dots, & \dots, & \dots, & \dots, \\
a_{p-1,0}, & a_{p-1,1}, & a_{p-1,2}, & a_{p-1,3}, & \dots, & a_{p-1,p-1},
\end{array}$$

each horizontal row and each vertical column being a system of connected letters.

If the horizontal rows are permuted among themselves, the group obtained, being primitive and of prime degree, must contain only substitutions of the form

$$(a_{k_1, k_2}, a_{mk_1+nk_2}),$$

the indices being taken modulo p .

The same will hold for the rows, which can yield only substitutions of the form

$$(a_{k_1, k_2}, a_{k_1, qk_2+r}).$$

Thus, finally, every substitution in H has the form

$$(a_{k_2, k_3}, a_{m_1 k_1 + n_1, m_2 k_2 + n_2}).$$

If a group G divides into n groups conjugate to the one just described, every substitution in G must transform into one another the cyclic substitutions of H , all written as follows:

$$(a_{k_1, k_2}, \dots, a_{k_1 + \alpha_1, k_2 + \alpha_2}, \dots). \quad (\text{a})$$

Suppose, then, that one substitution of G is formed by replacing, respectively,

$$\begin{array}{ll}
k_1 & \text{by } \phi_1(k_1, k_2), \\
k_2 & \text{by } \phi_2(k_1, k_2).
\end{array}$$

If in ϕ_1, ϕ_2 we substitute $k_1 + \alpha_1$ and $k_2 + \alpha_2$ for k_1, k_2 , the results must have the form

$$\phi_1 + \beta_1, \quad \phi_2 + \beta_2,$$

and from this it is easy to conclude immediately that all substitutions of G must be included in the formula

$$(a_{k_1, k_2}, a_{m_1 k_1 + n_1 k + \alpha_1 m_2 k_1 + n_2 k_2 + \alpha_2}), \quad (\text{A})$$

Now we know, from no. ^(*), that the substitutions of G can involve only $p^2 - 1$ or $p^2 - p$ letters. It cannot be $p^2 - p$, since in that case G would be nonprimitive. Thus, if in G we consider only the permutations in which, for example, the letter $a_{0,0}$ always retains the same place, we have only substitutions of order $p^2 - 1$ among the other $p^2 - 1$ letters.

But let us recall that it was merely for purposes of the proof that we supposed the primitive group G to divide into conjugate nonprimitive groups. Since this condition is by no means necessary, the groups will often be much more composite.

We must therefore determine when these groups can admit substitutions in which only $p^2 - p$ letters vary, and this investigation will occupy us for some time.

Let G be a group containing some substitution of order $p^2 - p$. I first assert that every substitution in this group is linear, that is, of form (A).

This is known to be true for substitutions of order $p^2 - 1$; it remains only to prove it for those of order $p^2 - p$. Consider, then, a group in which all the substitutions are m of order p^2 or order $p^2 - p$. (See the passage cited.)

The p letters that remain fixed in a substitution of order $p^2 - p$ must then be connected letters.

Suppose these connected letters are

$$a_{0,0}, \quad a_{0,1}, \quad a_{0,2}, \quad \dots, \quad a_{0,p-1}.$$

We may derive all substitutions in which these p letters do not change place from substitutions

^(*)This memoir follows a work of Galois that I do not possess; it is therefore impossible for me to identify the memoir cited here and below. (A. Ch.)

of the form

$$(a_{k_1, k_2}, a_{k_1, \phi k_2})$$

and from substitutions of order $p^2 - p$ whose period has p terms. (*See* again the passage cited.)

For the group to have the desired property, the former must necessarily reduce to the form

$$(a_{k_1, k_2}, a_{k_1, mk_2})$$

by what was seen for equations of degree p .

As for substitutions whose period has p terms, since they are conjugate to the preceding ones, we may suppose a group containing them but not the former. They must therefore transform the cyclic substitutions (a) into one another, and hence they too are linear.

We have thus reached the conclusion that the primitive permutation group on p^2 letters must contain only substitutions of form (A).

Now take the full group obtained by applying to the expression

$$a_{k_1, k_2}$$

all possible linear substitutions, and seek the subgroups of this group that can have the property required for solvability of equations.

What, first, is the total number of linear substitutions? It is clear that not every transformation of the form

$$k_1, k_2, m_1 k_1 + n_1 k_2 + \alpha_1, m_2 k_1 + n_2 k_2 + \alpha_2$$

is thereby a substitution; in a substitution, each letter of the first permutation must correspond to exactly one letter of the second, and conversely.

Thus, if we take any letter a_{l_1, l_2} of the second permutation and trace it back to the corresponding letter in the first, we must find a letter a_{k_1, k_2} whose indices k_1, k_2 are completely determined. Hence, for arbitrary l_1, l_2 , the two equations

$$m_1 k_1 + n_1 k_2 + \alpha_1 = l_1, \quad m_2 k_1 + n_2 k_2 + \alpha_2 = l_2,$$

must yield finite and determinate values of k_1, k_2 . The condition for such a transformation actually to be a substitution is therefore that $m_1n_2 - m_2n_1$ be neither zero nor divisible by the modulus p , which is the same thing.

I now assert that although, as we shall see, this linear substitution group does not always belong to equations solvable by radicals, it nevertheless has the following property: if any one of its substitutions has n fixed letters, then n divides the number of letters. Indeed, whatever the number of letters remaining fixed, this condition can be expressed by linear equations giving all the indices of one fixed letter in terms of some number of them. Giving each index that remains arbitrary its p values yields p^m systems of values, where m is some number. In the case before us, m is necessarily < 2 and is consequently 0 or 1. Therefore the number of substitutions cannot exceed

$$p^2(p^2 - 1)(p^2 - p).$$

Now consider only the linear substitutions in which $a_{0,0}$ does not move. If in this case we find the total number of permutations in the group containing all possible linear substitutions, it will suffice to multiply this number by p^2 .

First, by substituting p for the index k_2 , all substitutions of the form

$$(a_{k_1, k_2}, a_{m_1 k_1, k_2})$$

give $p - 1$ substitutions in all. We obtain $p^2 - p$ of them by adding the term $m_2 k_1$ to k_2 , as follows:

$$(k_1, k_2, m_1 k_1, m_2 k_1 + k_2). \quad (\text{m}')$$

On the other hand, it is easy to find a linear group of $p^2 - 1$ permutations such that in each substitution all letters except $a_{0,0}$ move. For, replacing the double index k_1, k_2 by the single index $k_1 + ik_2$,

where i is a primitive root of

$$x^{p^2-1} - 1 = 0 \pmod{p},$$

it is clear that every substitution of the form

$$[a_{k_1+k_2i}, a_{(m_1+m_2i)(k_1+k_2i)}]$$

is linear; but in these substitutions no letter remains in the same place, and there are $p^2 - 1$ of them.

We therefore have a system of $p^2 - 1$ permutations such that in each substitution all letters except $a_{0,0}$ move. Combining these substitutions with the $p^2 - p$ mentioned above gives

$$(p^2 - 1)(p^2 - p) \text{ substitutions.}$$

Now we saw *a priori* that the number of substitutions in which $a_{0,0}$ remains fixed could not exceed $(p^2 - 1)(p^2 - p)$. It is therefore exactly $(p^2 - 1)(p^2 - p)$, and the full linear group has altogether

$$p^2(p^2 - 1)(p^2 - p) \text{ permutations.}$$

It remains to find the subgroups of this group having the property required for solvability by radicals. For this purpose we shall make a transformation aimed at lowering as far as possible the general equations of degree p^2 whose group is linear.

First, since the cyclic substitutions of such a group are transformed into one another by every other substitution of the group, we may lower the degree of the equation by one and consider an equation of degree $p^2 - 1$ whose group has only substitutions of the form

$$(b_{k_1, k_2}, b_{m_1 k_1 + n_1 k_2}, m_2 k_1 + n_2 k_2),$$

the $p^2 - 1$ letters being

$$\begin{array}{cccccc} & b_{0,1}, & b_{0,2}, & b_{0,3}, & \dots, & \\ b_{1,0}, & b_{1,1}, & b_{1,2}, & b_{1,3}, & \dots, & \\ b_{2,0}, & b_{2,1}, & b_{2,2}, & b_{2,3}, & \dots, & \\ \dots, & \dots, & \dots, & \dots, & \dots, & \end{array}$$

I now observe that this group is nonprimitive, so all letters for which the ratio of the two indices is the same are connected letters. Replacing each system of connected letters by a single letter gives a group whose substitutions all have the form

$$\left(b_{\frac{k_1}{k_2}}, b_{\frac{m_1 k_1 + n_1 k_2}{m_2 k_1 + n_2 k_2}} \right),$$

where $\frac{k_1}{k_2}$ are the new indices. Replacing this ratio by one index k , we see that the $p+1$ letters are

$$b_0, \quad b_1, \quad b_2, \quad b_3, \quad \dots, \quad b_{p-1}, \quad b_{\frac{1}{0}},$$

and the substitutions have the form

$$\left(k, \frac{mk+n}{rk+s} \right).$$

Let us determine how many letters remain fixed in each of these substitutions; for this we must solve the equation

$$(rk+s)k - m(mk+n) = 0,$$

which has two, one, or no roots according as $(m-s)^2 + 4nr$ is a quadratic residue, zero, or a quadratic nonresidue. In these three cases respectively, the substitution has order $p-1$, p , or $p+1$.

As a type for the first two cases we may take substitutions of the form

$$(k, mk+n),$$

in which only the letter $b_{\frac{1}{0}}$ does not move. It follows that the total number of substitutions in the reduced group is

$$(p+1)p(p-1).$$

Having reduced this group in this way, we shall now treat it generally. We shall first determine when a subgroup containing substitutions of order p can belong to an equation solvable by radicals.

In this case the equation would be primitive and could not be

solvable by radicals unless $p + 1 = 2^n$ for some n .

We may suppose that the group contains only substitutions of order p and order $p + 1$. Consequently all substitutions of order $p + 1$ are similar and have a period of two terms.

Take the expression

$$\left(k, \frac{mk + n}{rk + s}\right),$$

and determine when this substitution can have a period of two terms. For this, the inverse substitution must coincide with it. The inverse substitution is

$$\left(k, \frac{-sk + n}{rk - m}\right).$$

Thus $m = -s$, and all substitutions in question are

$$\left(k, \frac{mk + n}{k - m}\right),$$

or equivalently

$$k, m + \frac{N}{k - m},$$

where N is some number, the same for every substitution, since these substitutions must be transformed into one another by all substitutions of order p , $(k, k + m)$. Moreover, these substitutions must be conjugate to one another. Thus, if

$$\left(k, m + \frac{N}{k - m}\right), \quad \left(k, n + \frac{N}{k - n}\right)$$

are two such substitutions, we must have

$$n + \frac{N}{\frac{N}{k - m} + m - n} = m + \frac{N}{\frac{N}{k - n} + n - m},$$

namely,

$$(m - n)^2 = 2N.$$

Thus the difference between two values of m can assume

only two distinct values; hence m can have no more than three values, and therefore $p = 3$. Thus only in this case can the reduced group contain substitutions of order p .

Indeed, the reduced equation is then of the fourth degree and is consequently solvable by radicals.

We therefore know that, in general, the substitutions of our reduced group must include none of order p . Can there be any of order $p - 1$? This is what I shall investigate⁽¹⁾.

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END.

⁽¹⁾I searched Galois's papers in vain for the continuation of what has just been read. (A. Ch.)

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